Papers read

An image is worth 16x16 words: Transformers for image recognition at scale — https://arxiv.org/pdf/2010.11929/100

Mambavision: A hybrid mamba-transformer vision backbone — https://openaccess.thecvf.com/content/CVPR2025/papers/Hatamizadeh_MambaVision_A_Hybrid_Mamba-Transformer_Vision_Backbone_CVPR_2025_paper.pdf

Swin transformer: Hierarchical vision transformer using shifted windows — https://openaccess.thecvf.com/content/ICCV2021/papers/Liu_Swin_Transformer_Hierarchical_Vision_Transformer_Using_Shifted_Windows_ICCV_2021_paper.pdf

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Swin Transformer Variants on HAM10000 Skin Lesion Dataset

HAM10000 (Human Against Machine with 10000 training images) is a skin lesion classification dataset consisting of **10,015 dermatoscopic images** across **7 classes**.

Data split

Train

after oversampling

Validation

validation set size

Test

test set size

Handling class imbalance

Oversampling

Applied on the training set to reduce class imbalance.

Class-weighted loss

Used during training to give minority classes more emphasis.

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Processing and augmentation

The input images were preprocessed and augmented to improve model robustness and reduce overfitting.

Preprocessing

Images were resized and normalized before training.

Rotation

Random rotations helped the model handle different lesion orientations.

Flipping

Horizontal and vertical flips increased variation in the training set.

Color jittering

Brightness, contrast, and saturation adjustments improved generalization.

Additional augmentation

Optional shifts, zooms, and other mild transforms further expanded the dataset.

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Architecture Overview

Swin Transformer is a hierarchical vision transformer that performs self attention within local windows instead of globally, making it computationally efficient.

Hierarchical design

Features are built across multiple stages for multi-scale representation.

Variant	Embed Dim	Depths	Heads	Params
Swin-T	96	[2, 2, 6, 2]	[3, 6, 12, 24]	28M
Swin-S	96	[2, 2, 18, 2]	[3, 6, 12, 24]	50M
Swin-B	128	[2, 2, 18, 2]	[4, 8, 16, 32]	88M

All three were implemented from scratch – no pretrained weights used.

Key distinction: Swin Transformer uses local window attention, while standard vision transformers use global attention across all patches.

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Training setup:

Setting	Value
Optimizer	AdamW
Learning rate	1e-4
Weight decay	0.05
Scheduler	CosineAnnealingLR (T_max=100)
Loss	CrossEntropyLoss (class weighted)
Batch size	32
Epochs	100
Hardware	Kaggle T4 x 2

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Results:

MODEL	PARAMS	BEST VAL ACC	TEST ACC	TRAIN TIME
swin-t	28M	77.20%(ep87)	74.65%	~3.5hrs
swin-s	50M	77.20%(ep87)	74.55%	~5hrs
swin-b	88M	75.87%(ep60)	73.05%	~7.5hrs

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Key observations

Model Size vs Accuracy

- Swin-t achieved the highest test accuracy, while Swin-b achieved the lowest.
- There is an inverse relationship between model size and performance.
- A bigger model did not help on this small medical dataset.

Convergence Speed

- Swin-t reached best val acc at ep87, Swin-s at ep87, and Swin-b at ep60.
- Swin-t took longer to converge but generalized better.
- Swin-s peaked early but plateaued.

Training Efficiency

- Swin-t: 28M params, ~3.5hrs
- Swin-s: 50M params, ~5hrs
- Swin-b: 88M params, ~7.5hrs

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Training Efficiency

Swin-t

74.65% in **3.5 hrs**

Best accuracy per GPU hour

Swin-s

74.55% in **5 hrs**

Middle ground on speed and accuracy

Swin-b

73.05% in **7.5 hrs**

Most expensive yet **least accurate**

Overfitting Analysis

Swin-T

16.23%

Smallest gap between training and validation accuracy

90.88 - 74.65

Swin-S

17.95%

Moderate gap between training and validation accuracy

92.50 - 74.55

Swin-B

18.41%

Largest gap and strongest sign of overfitting

91.46 - 73.05

Swin-t performed best overall, while **Swin-b** performed worst and showed the most overfitting.

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Why Overfitting Increased

Too many parameters

Swin-B had **88M params** for just **10,827 images**, making it easier to memorize training data.

Oversampling duplicated data

Oversampling created repeated minority-class images instead of adding new examples.

No pretrained weights

Random initialization required more data to learn robust general features.

Augmentation was not enough

Data augmentation alone could not close the capacity gap in the larger models.

Lower-capacity models generalized better

Swin-T was forced to learn transferable features instead of memorizing patterns.

Limitations

- No pretrained weights → trained from scratch
- Small dataset → only 10,000 images

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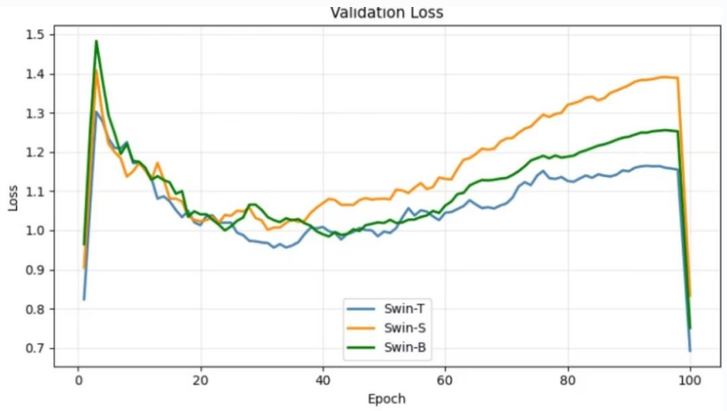
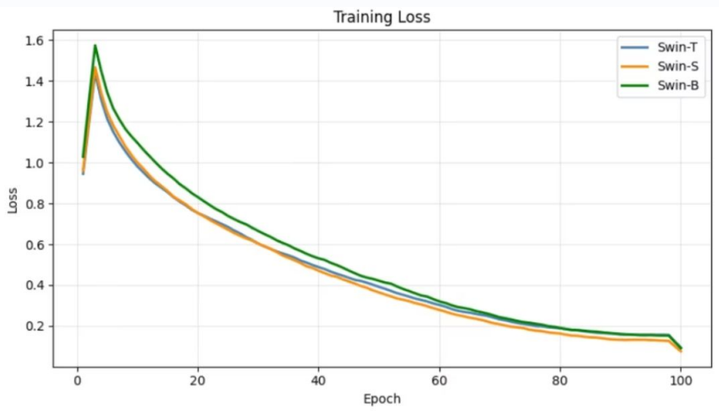
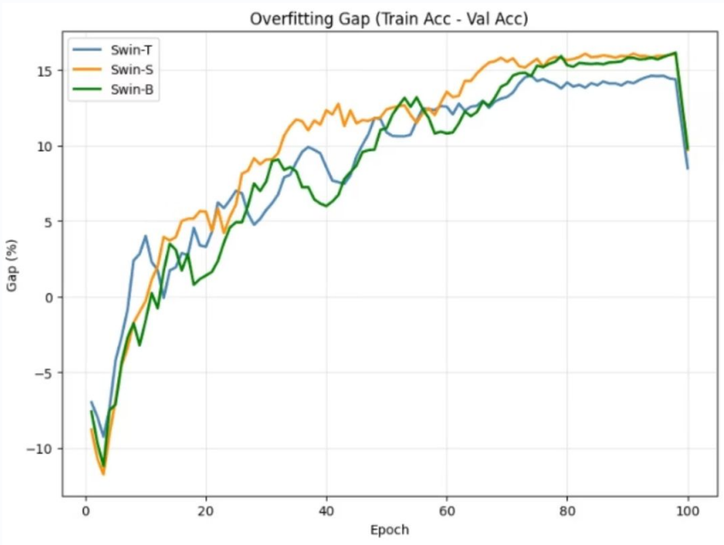
Plots: (w/ smoothing function applied)







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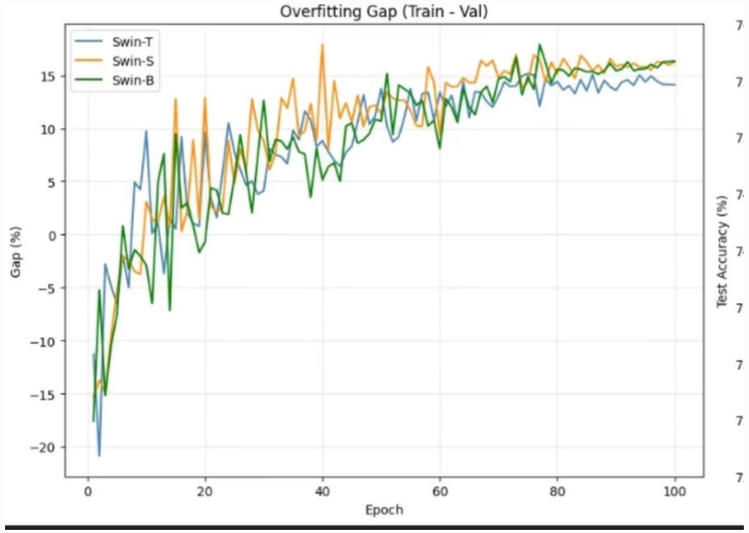
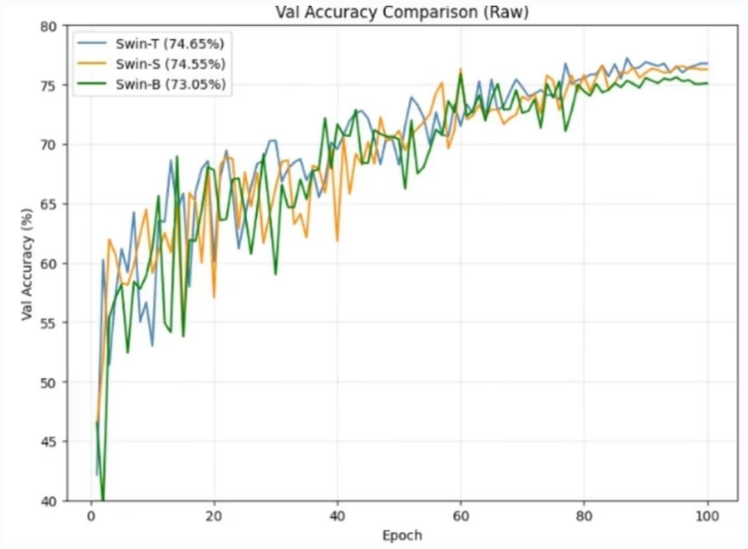
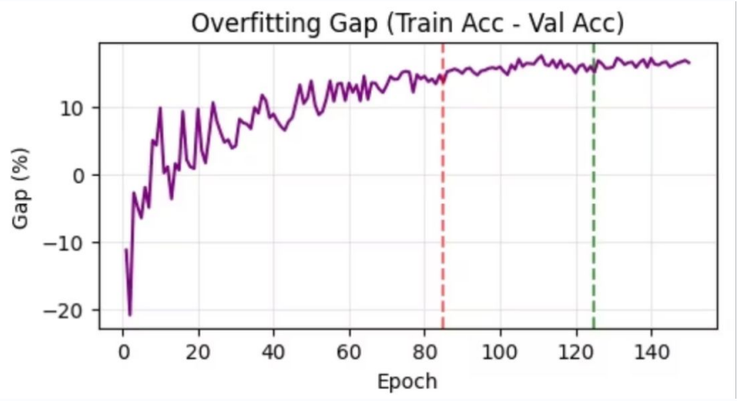
Plots:

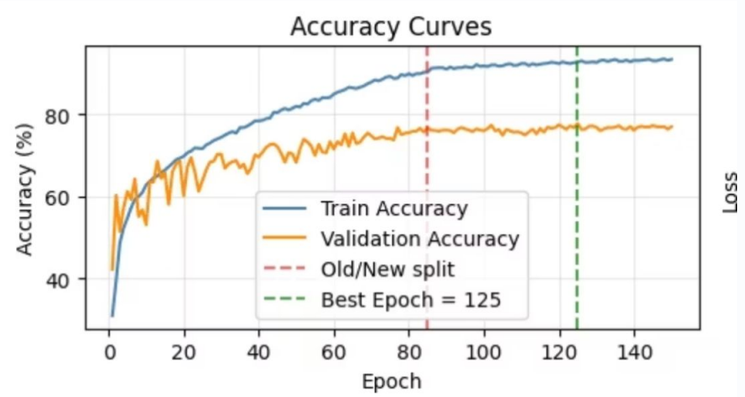
100 epochs





150 epochs





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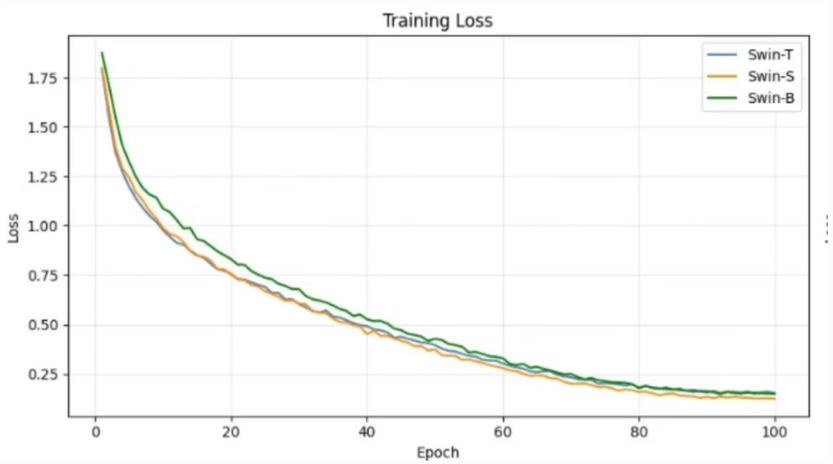
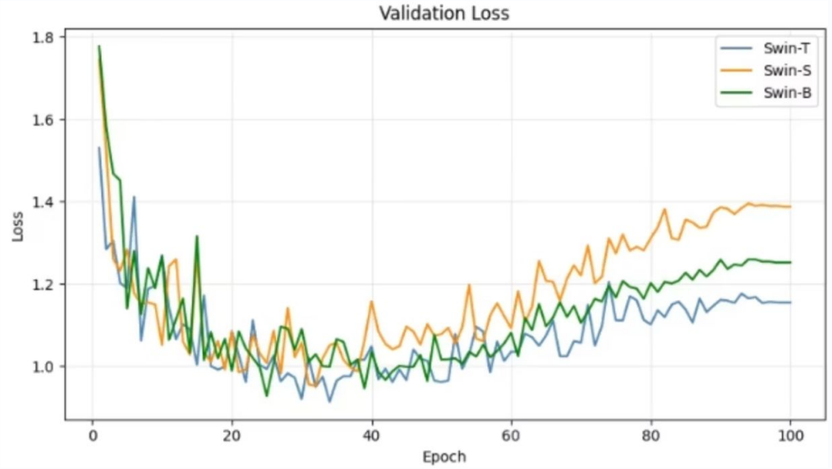
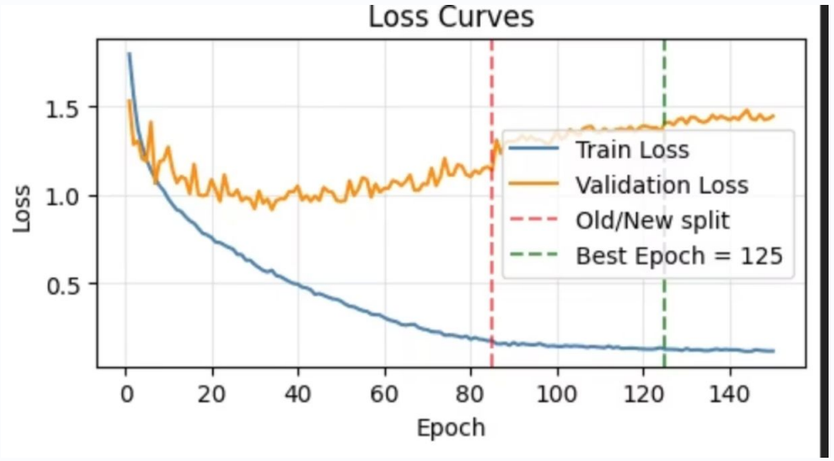
100 epochs





150 epochs



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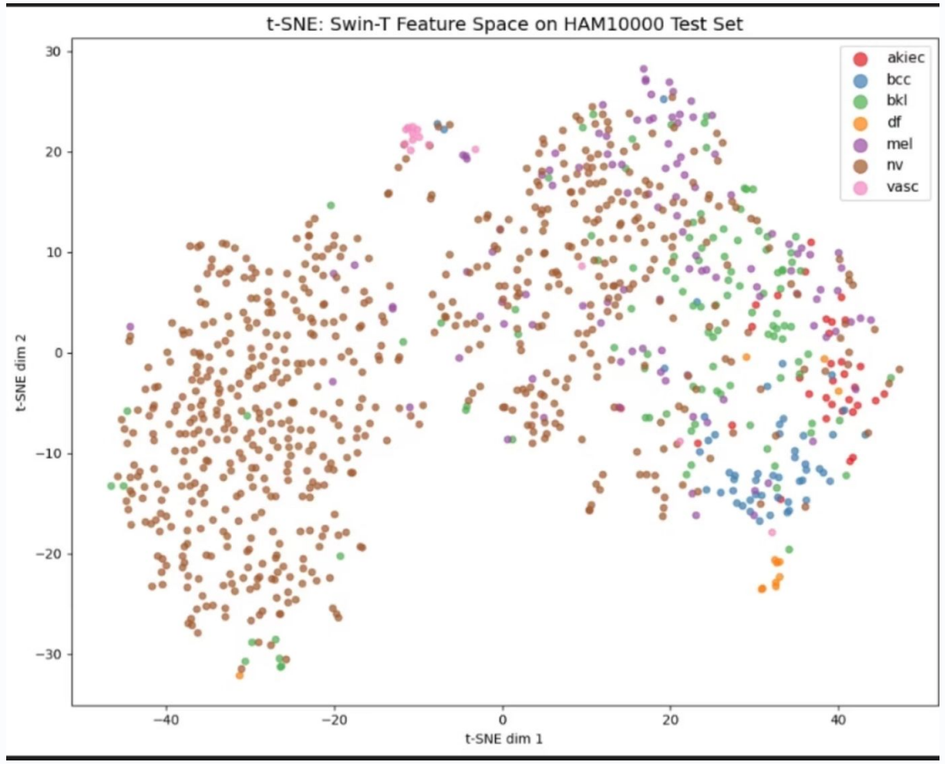
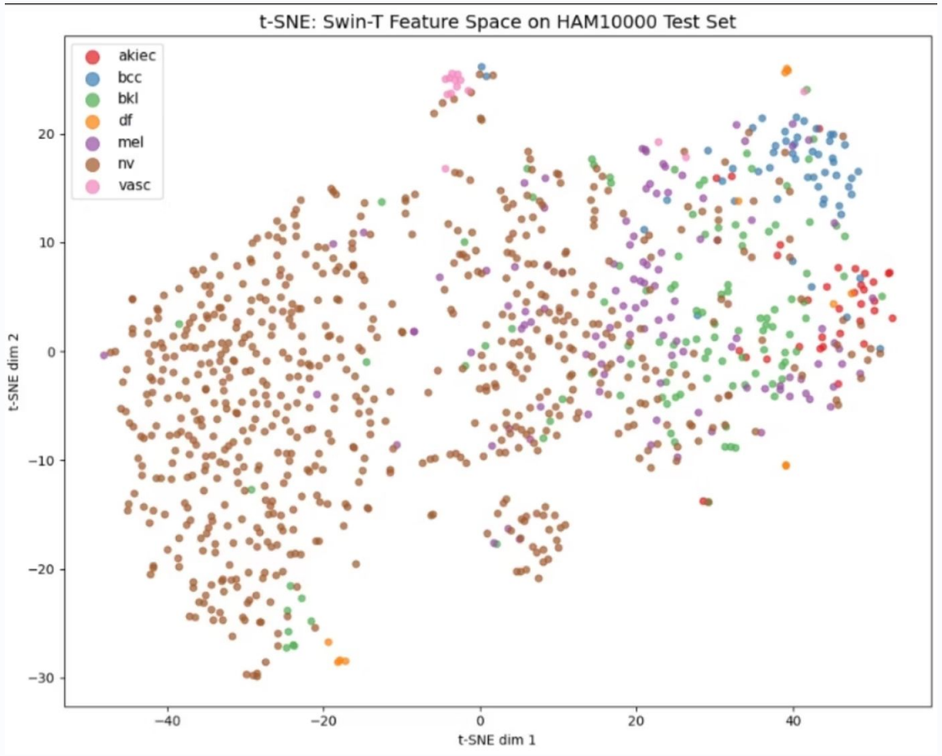
t-sne plot

100 epochs



150 epochs



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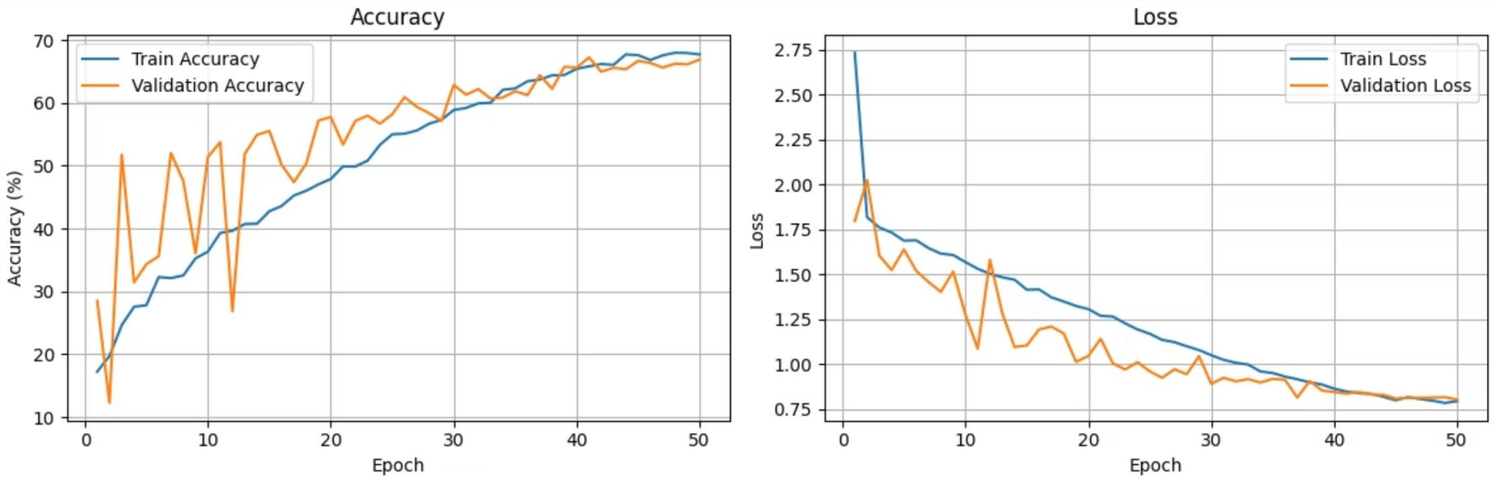
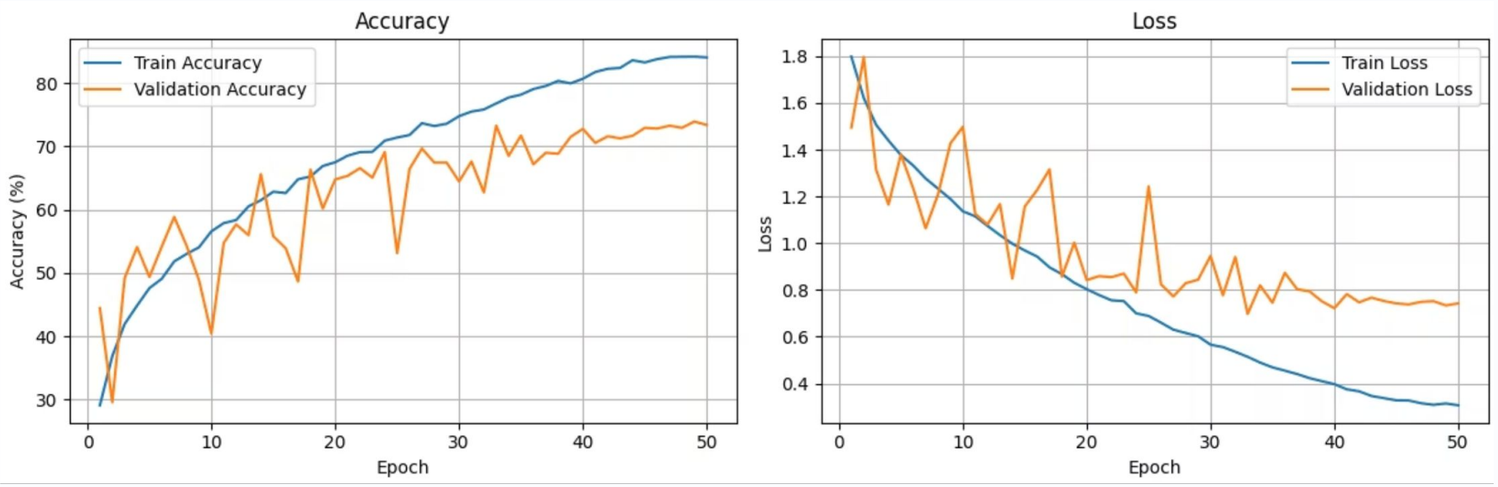
cnn 50 epochs (test acc-64.47%)



ResNet18 50 epochs (test acc-71.86%)



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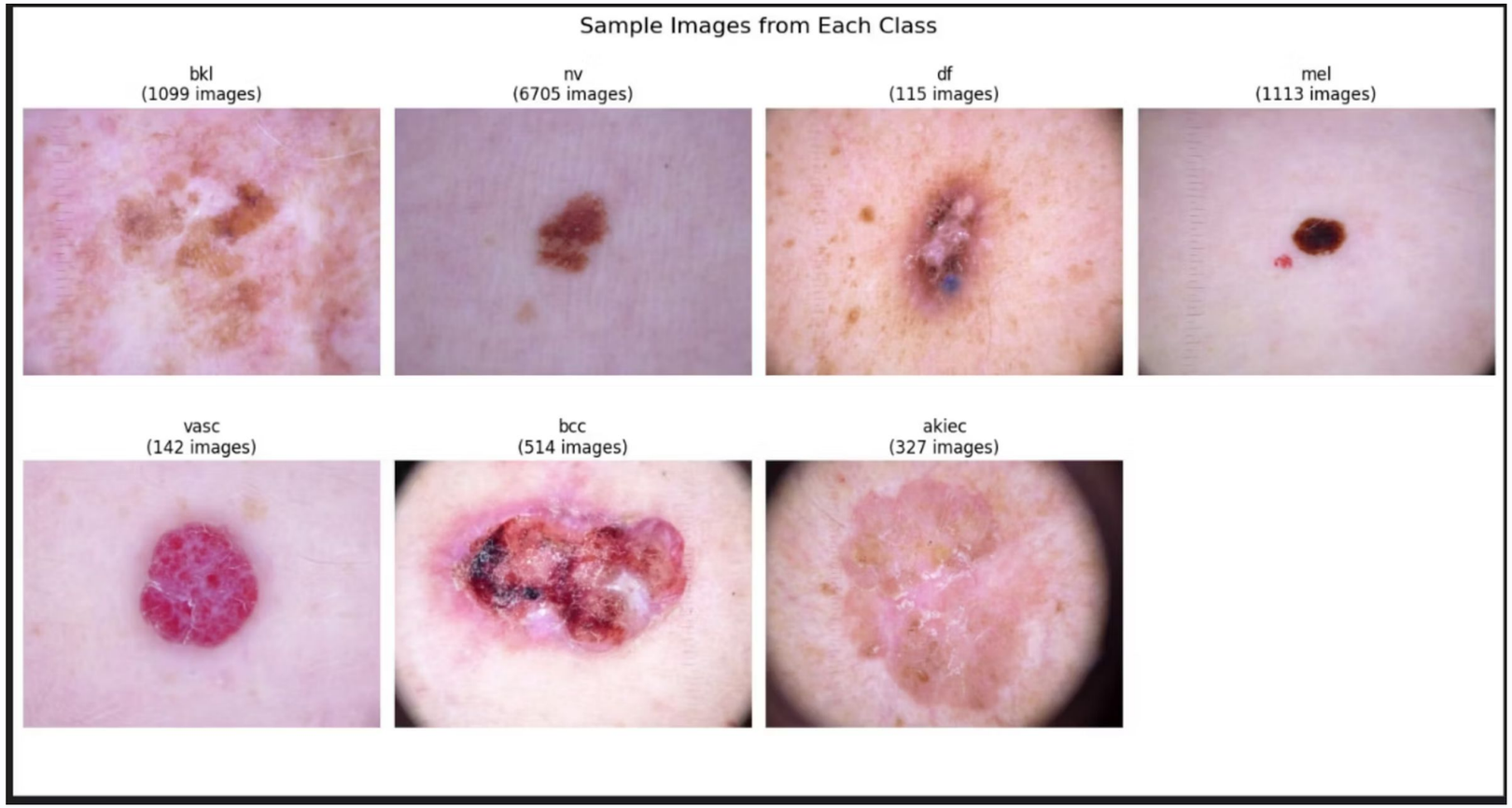
1. nv(brown) - mel(purple) most similar
2. nv(brown) - bkl(green)
3. akie(red) - bkl(green)
4. bcc(blue) - mel(purple)

Mel, bcc, akiec - cancerous

Nv, bkl, df, vasc - non cancerous



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Swin-T (100 epochs)

First run: 74.65% test acc

Best val acc: epoch 87

Second run: 75.95% test acc

Best val acc: epoch 85

Swin-T (150 epochs)

First run: 76.85% test acc

Best val acc: epoch 125

Second run: 77.15% test acc

Best val acc: epoch 146

Early Stopping Analysis

Patience-based early stopping on validation accuracy

Patience 10: early stopping at epoch 14

61.88% test acc

Patience 20: early stopping at epoch 28

65.27% test acc

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ISIC2019 dataset

Dataset

- 8 classes: MEL, NV, BCC, AK, BKL, DF, VASC, SCC
- Strong class imbalance (NV dominant)
- ~17k train, 3.8k val, 3.8k test

Challenges

- Noisy images (hair, artifacts, illumination variations)
- Harder than HAM10000

Preprocessing & Training

- Resize: 224 x 224
- Data augmentation
- Weighted CrossEntropy Loss

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Results

Dataset	Model	Test accuracy
HAM10000	CNN	64.47%
HAM10000	ResNet18	71.86%
HAM10000	Swin-T	74.76%
ISIC2019	CNN	54.97%
ISIC2019	ResNet18	61.76%
ISIC2019	Swin-T	61.66%
ISIC2019	DINOv2 (fine-tuned)	87.32%
ISIC2019	MambaVision-T (fine-tuned)	71.15%

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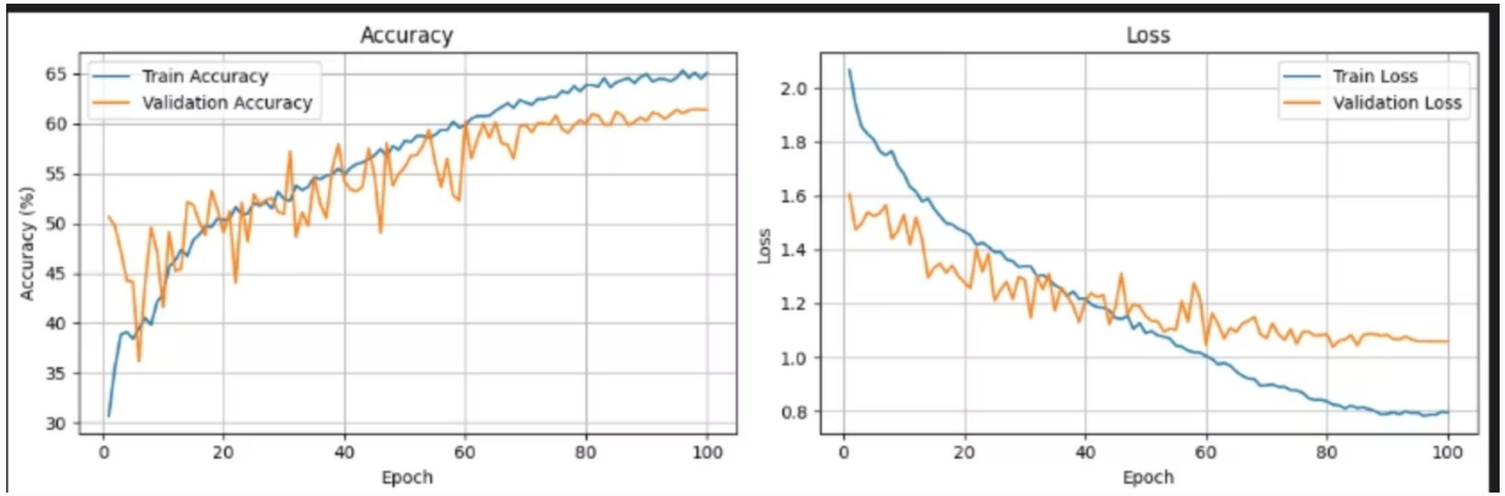
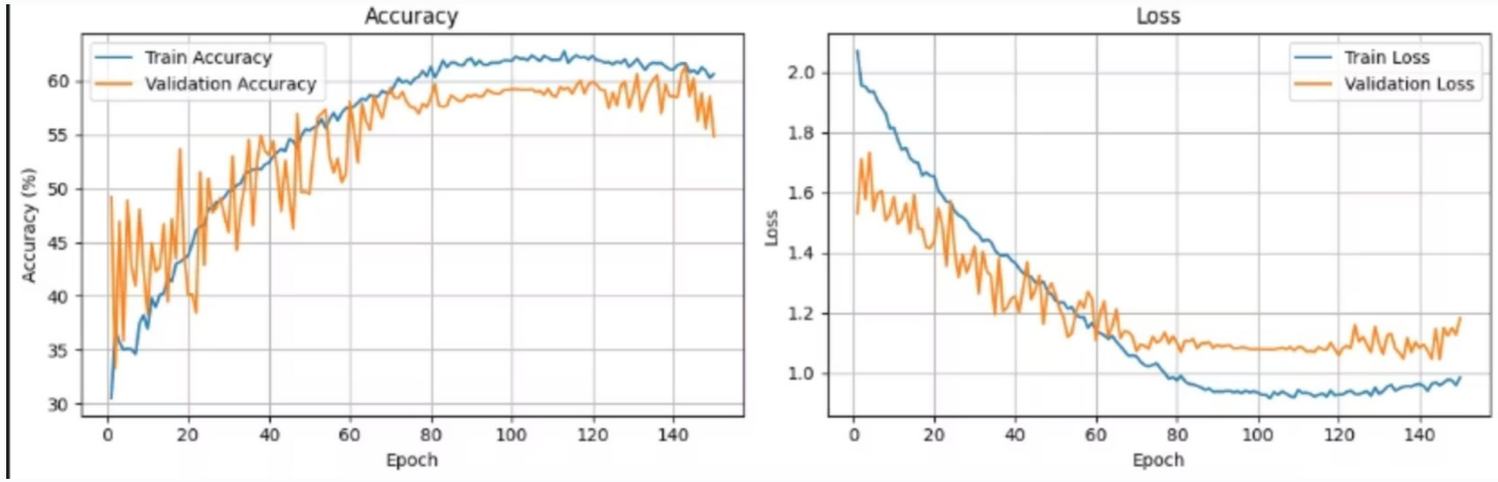
swin-t 100 epochs (test acc- 61.66%)



swin-t 150 epochs (test acc- 55.29%)



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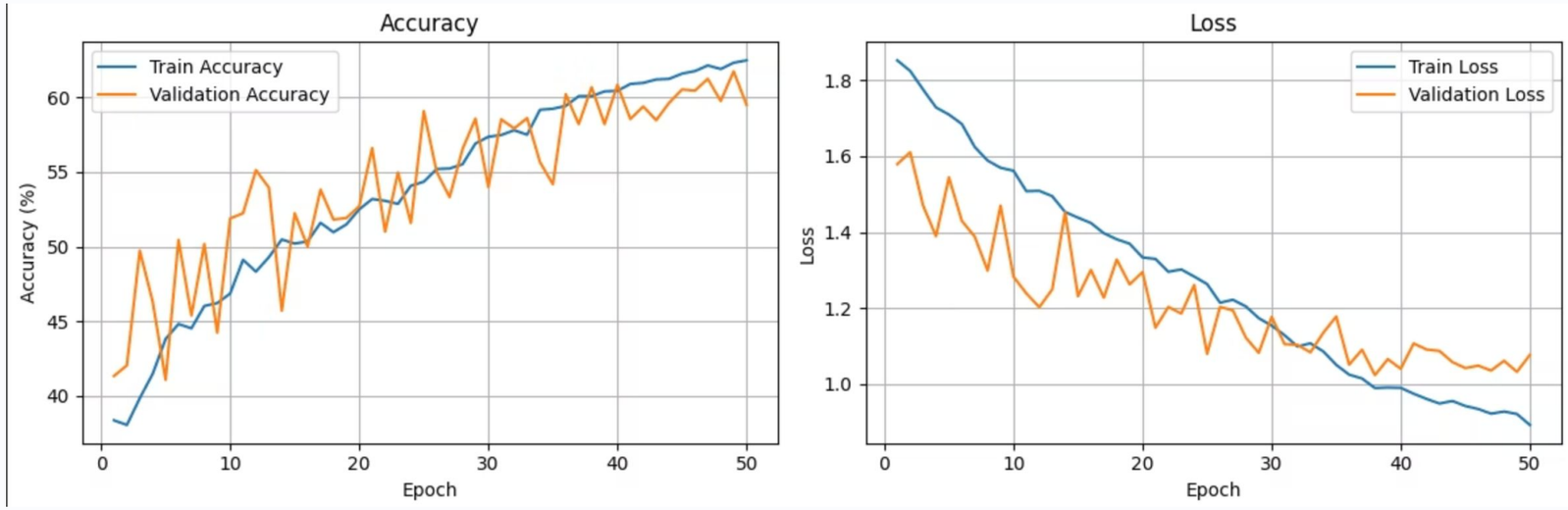
ResNet18 50 epochs (test acc- 61.76%)



More stable performance and better generalization than Swin-T

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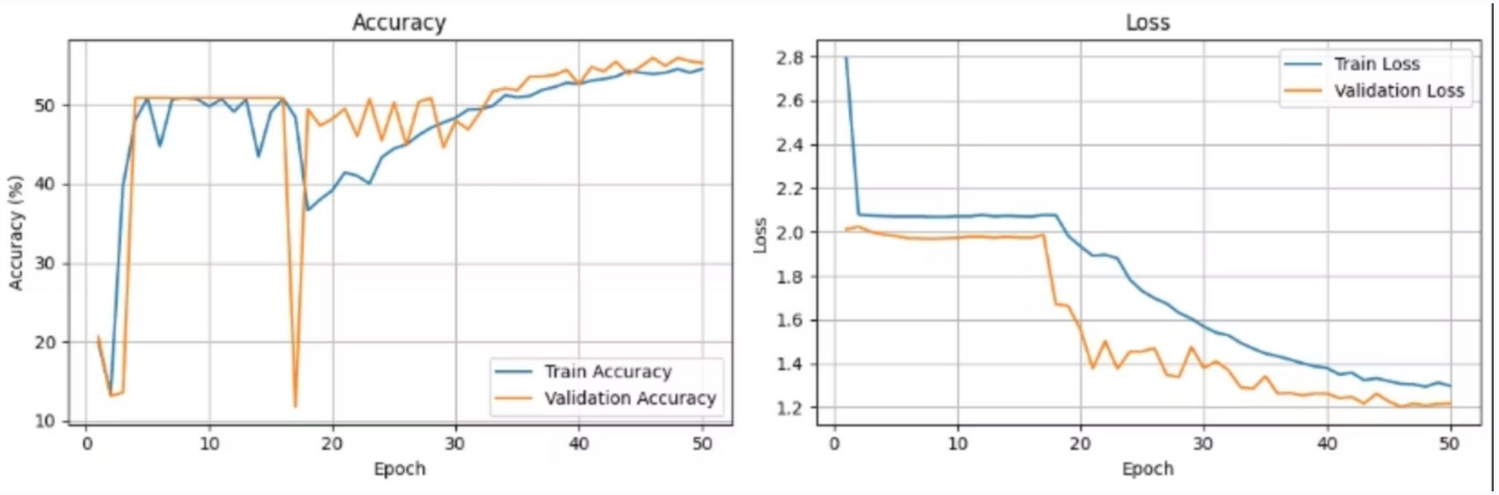
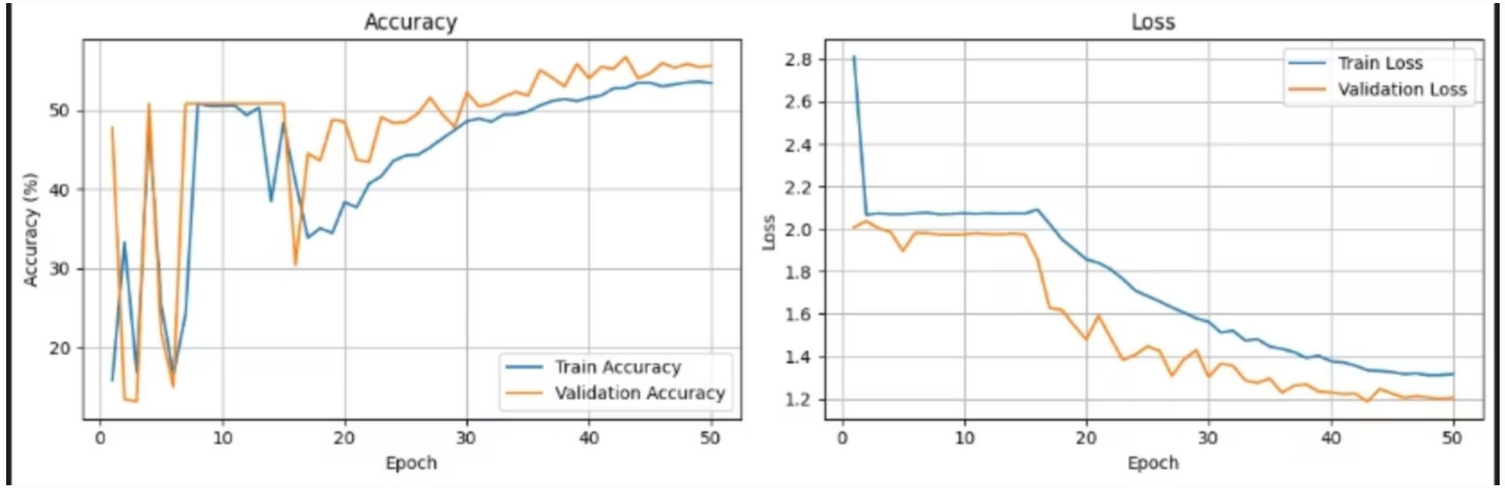
cnn 50 epochs(first run) (test acc-54.97%)



cnn 50 epochs (seocnd run) (test acc-56.37%)



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key takeaways

- Model performance depends strongly on dataset characteristics
- swin performs well on clean datasets(ham10000) but struggles on noisy data(ISIC)
- resnet18 shows more stable generalization across datasets
- higher model capacity can lead to overfitting on small/imbalanced datasets

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Training setup:

Model setup

- Pretrained models: DINOv2, MambaVision-Tiny
- Modified classification head for 8 classes

Training strategy

- Phased fine-tuning
  - Phase 1: Train classification head
  - Phase 2: Unfreeze deeper layers
  - Phase 3: Full model fine-tuning

Optimization

- Optimizer: AdamW
- Learning rate:
  - Head: higher LR
  - Backbone: lower LR
- Scheduler: Cosine Annealing

Regularization Techniques

- Mixup augmentation (applied with 50% probability)
- Exponential Moving Average(EMA)
- Test-Time Augmenation(TTA)

Input Configuration

- Image resolution: 224 for mambavision-t, 384 for dinov2
- Batch size: 32 for mambavision-t, 8 for dinov2

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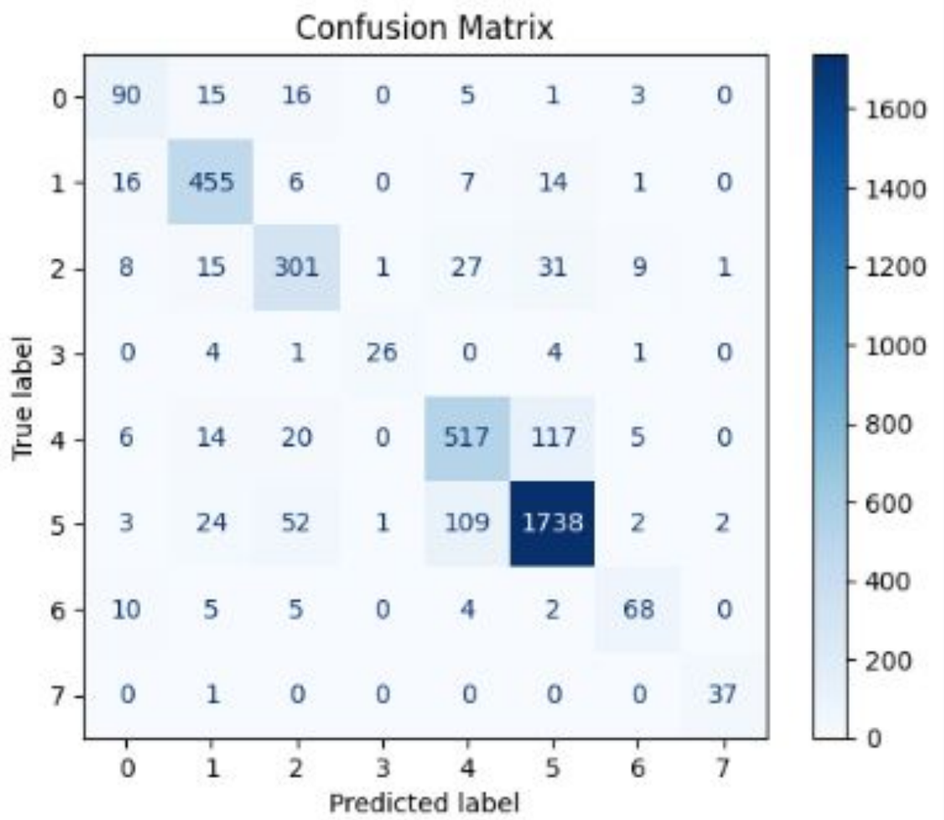
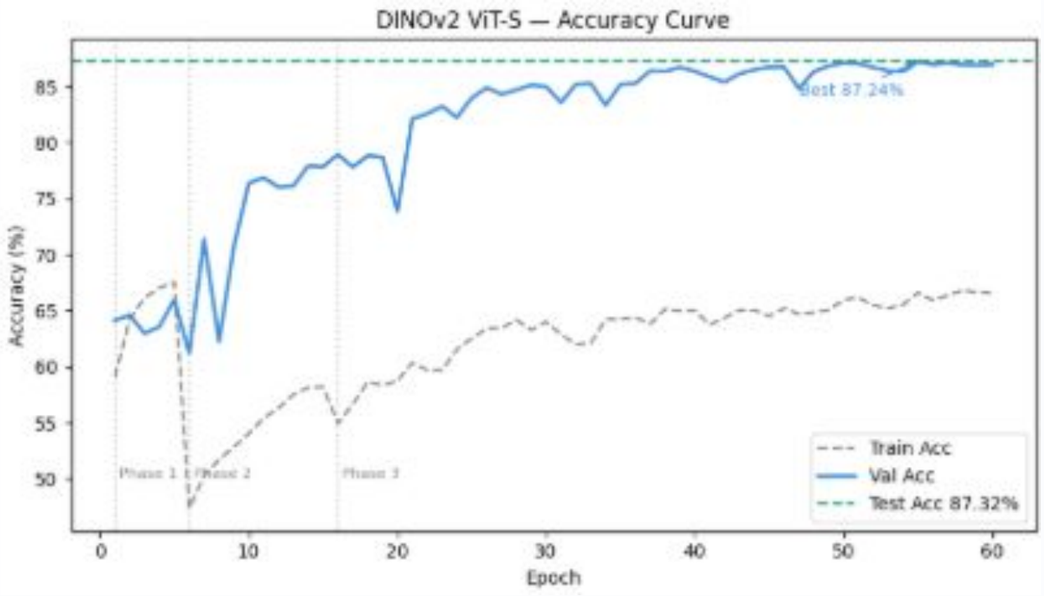
Results- DINOv2





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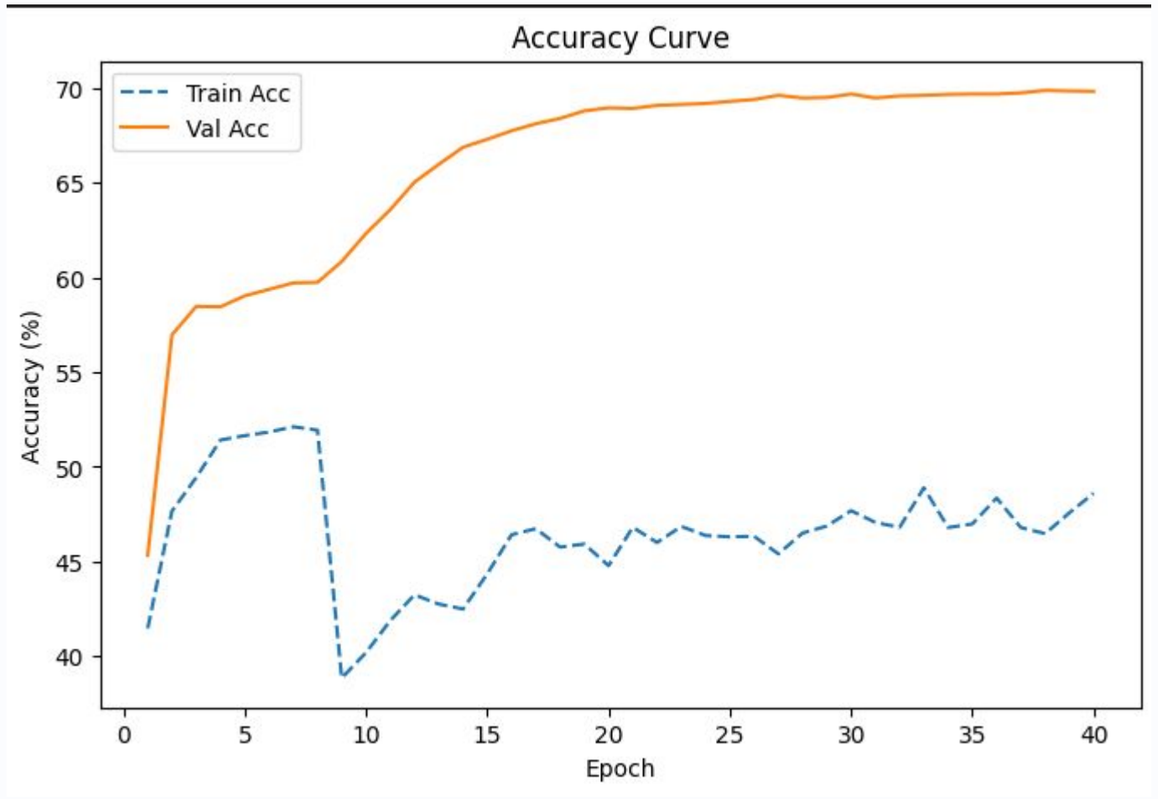

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